

Topological HDDM Formulation: Synthesizing Cerebellar Topology and Stochastic Decision Dynamics

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This document serves as the formal mathematical consensus bridging the continuous, high-dimensional geometry of the cerebellar manifold with the stochastic accumulator dynamics of the cortex. It resolves the strict sub-optimality of static decision boundaries by introducing a topologically-gated dynamic policy.

1. The Computable Formulation: System Equations

The dynamic policy $\pi_{\Omega} = (v_t, a_t)$ modulates the cortical evidence accumulator by extracting the topological uncertainty of the cerebellar network and mapping it through biophysical neuromodulatory cascades.

A. Homologically-Filtered Topological Uncertainty ($\tilde{\Omega}_t$)

To preclude stochastic artifacts from misclassifying sub-manifolds, we filter the persistence module using a Wasserstein distance threshold θ_W . Let $\mathcal{P}_t$ be the persistence diagram of the active Purkinje population state Z_t over a temporal window. The filtered Symplectic Persistent Entropy $\tilde{\Omega}_t$ is defined as:

$$\tilde{\Omega}_t = c(Z_t) \cdot \mathbb{I}_{\{W(\mathcal{P}_t, \emptyset) > \theta_W\}} \left(- \sum_{j \in \mathcal{P}_{\text{filtered}}} p_j \log p_j \right)$$

where $c(Z_t)$ is the symplectic capacity of the active region, and p_j represents the normalized lifetimes of homological features strictly exceeding the noise threshold.

B. Biologically-Smoothed Neuromodulatory Surge (N_t, eff)

To protect classical Hamilton-Jacobi-Bellman (HJB) structures from discontinuous Deep Cerebellar Nuclei (DCN) Dirac impulses (rebound bursting), the effective noradrenergic gating N_t, eff integrates extracellular clearance and reuptake kinetics via convolution:

$$N_{\text{raw}}(t) = \Gamma \left[1 + \exp \left(\frac{1}{k_h} \left(\frac{g_L}{E_L + g_{\text{GABA}}} e^{-\kappa \tilde{\Omega}_t} E_{\text{Cl}} \right) \left(\frac{g_L}{E_L + g_{\text{GABA}}} e^{-\kappa \tilde{\Omega}_t} - V_{1/2} \right) \right) \right]^{-1}$$

$$N_t, \text{eff} = (N_{\text{raw}} * h_{\text{NA}})(t) = \int_0^t N_{\text{raw}}(s) e^{-\frac{t-s}{\tau_{\text{NA}}}} ds$$

where τ_{NA} defines the reuptake time constant, ensuring N_t, eff is strictly C^1 .

C. The Dynamically Gated Policy (π_{Ω})

The drift rate v_t and decision boundary a_t of the Stochastic Delay Differential Equation (SDDE) are continuously modulated by the smoothed biophysical state:

$$\begin{aligned} v_t(N_t, \text{eff}) &= v_{\text{base}} + \eta_v N_t, \text{eff} \\ a_t(N_t, \text{eff}) &= a_{\text{base}} e^{-\eta_a N_t, \text{eff}} \end{aligned}$$

2. Biophysical Mapping

The abstraction maps strictly to physical cerebellar microcircuitry without requiring non-biological mechanisms like Backpropagation Through Time (BPTT).

- Topological Synchrony:** As the network enters a low-dimensional heuristic state ($\mathcal{M}_H$), entropy vanishes ($\tilde{\Omega}_t \rightarrow 0$), driving Purkinje population synchrony $S_t \propto e^{-\kappa \tilde{\Omega}_t} \rightarrow 1$.
 - DCN Hyperpolarization & Calcium Priming:** Massive synchronous GABAergic inhibition drives the DCN membrane voltage $V_{\text{DCN}} \rightarrow E_{\text{Cl}}$. This profound hyperpolarization is required to de-inactivate T-type Calcium channels ($h_{\infty} \rightarrow 1$ as $V \rightarrow V_{1/2}$).
 - LC Rebound Bursting:** Upon release of inhibition, the primed Calcium channels trigger a massive rebound burst, projecting to the Locus Coeruleus (LC) and releasing a noradrenaline surge $N_{\text{raw}} \rightarrow \Gamma$.
 - Cortical Gating:** The subsequent surge in effective noradrenaline N_t, eff drives the boundary a_t toward 0 and spikes the drift rate v_t , enforcing a rapid, reactive choice (Heuristic Execution). Conversely, during high-entropy deliberation ($\mathcal{M}_D$), *asynchrony prevents the rebound burst, keeping $N_t, \text{eff} \approx 0$* , thereby preserving a wide boundary a_{base} for slow evidence accumulation.
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3. The Unifying Theorem: Dynamic Optimality via Viscosity Solutions

Theorem: Let the behavioral state space manifold be $\mathcal{M} = \mathcal{M}_H \cup \mathcal{M}_D$, subject to discontinuous capacity transitions in $\mathcal{C}(Z, t)$. The optimal value function $V(x, t)$ of the generalized reward rate functional $\mathcal{U}(\pi)$ under the dynamically gated policy $\pi_{\Omega} = (v_t(N_t, \text{eff}), a_t(N_t, \text{eff}))$ is the unique, continuous **viscosity solution** to the HJB variational inequality:

$$\min \left\{ -\frac{\partial V}{\partial t} - \sup_{v, a} \left[v_t(N_t, \text{eff}) \frac{\partial V}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2 V}{\partial x^2} - c \right], V(x, t) - B(x, t) \right\} = 0$$

where $B(x, t)$ maps the boundary conditions at $|x| \geq a_t(N_t, \text{eff})$.

Proof Summary: Because $\tilde{\Omega}_t$ incorporates discontinuous jumps $\mathcal{C}(Z, t)$ during discrete manifold state transitions, classical $\mathcal{C}^{2,1}$ solutions fail at the boundaries of $\mathcal{M}_H \cap \mathcal{M}_D$. However, the extracellular smoothing convolution

$N(t, \text{eff})$ strictly bounds the temporal derivatives of the policy parameters a_t and v_t .

Consequently, for any smooth test function $\phi \in C^\infty$ such that $V - \phi$ attains a local extrema in the continuation region, the sub-differentials satisfy the classical HJB differential operator. Therefore, V uniquely satisfies the Crandall-Lions viscosity definition, continuously resolving the maximal generalized reward rate unconditionally across all non-differentiable manifold boundaries.

Because the dynamic policy strictly avoids the suboptimal continuation intrinsic to a rigid static boundary C (which forces wasted time in $\mathcal{M}_H$ and premature stopping in $\mathcal{M}_D$), we conclude: $\mathcal{U}(\pi_\Omega) > \mathcal{U}(\pi_0) \quad \blacksquare$